DolmaNet: Truthful Mechanism Learning with Convex Utility Potentials

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Abstract

We introduce DolmaNet, a neural architecture for truthful mechanism learning with additive bidders. It represents each bidder's indirect utility by a deep convex potential conditioned on rival reports; allocations are own-report gradients and payments follow the envelope identity. A sign-aware normalization bounds allocation coordinates in $[0, 1]$ without discarding signed input paths, making bidderwise dominant-strategy incentive compatibility and ex-post individual rationality structural. For the increasing-capacity, zero-margin class, we construct uniform approximations of continuous truthful utilities compatible with the chosen context representation, with allocations and payments converging in expected error under absolutely continuous type distributions. When a target is unit-supply feasible, the approximating sequence also has vanishing expected positive supply excess. Individual neural mechanisms, however, are not guaranteed to satisfy joint unit supply.

We evaluate DolmaNet on binary-valued auctions, reproducible-goods environments, and unit-supply settings using finite-menu controls and mechanism-level diagnostics. The experiments examine the gap between representational capacity and learned performance through allocation and payment geometry. Results on reproducible goods are competitive with published baselines in several settings, while unit-supply experiments assess revenue together with mean excess and profile-level violations. These evaluations separate structural incentive guarantees from the challenges of optimization and shared-resource feasibility.

CCS Concepts

• **Theory of computation → Algorithmic mechanism design**.

Keywords

automated mechanism design, auctions, dominant-strategy incentive compatibility, convex neural networks, differentiable economics

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1 Introduction

Truthful mechanism learning can parameterize either a bidder's choices or the utility that those choices attain. RegretNet learns allocation and payment rules with structural supply feasibility and estimated-regret penalties [4]. GemNet learns rival-conditioned finite menus and transforms their prices to enforce joint compatibility [10]. TEDI instead learns continuous prices over a product outcome space and computes outcomes by utility maximization [7]. We study a complementary implementation: represent indirect utility directly, so that differentiation selects an implicit menu option without a separate outcome-optimization problem.

For bidder i, DolmaNet uses

$$U_i(b_i, b_{-i}) = \Phi_\theta(b_i, C_i(b_{-i})) - \Phi_\theta(0, C_i(b_{-i})),$$

where the context contains only rival reports and Φ_θ is convex in the own report. Allocations are own-report gradients and payments follow the envelope identity. This utility representation is classical and already used by RochetNet [4, 9]; conditional convex approximation and gradient-based mechanism learning are also established [6, 8]. Our contribution is a deep rival-conditioned parameterization whose signed input paths can be normalized to bound allocations globally without destroying its approximation property.

Convexity alone does not bound a gradient in $[0, 1]^m$. DolmaNet propagates signed derivative intervals and normalizes paths *inside the potential*, before differentiation. This preserves convexity and the allocation–payment identity. Independent bidder evaluations admit vectorization, while training differentiates through the resulting allocation gradients.

The outcome space matters. DolmaNet is a truthful mechanism on $\prod_i [0, 1]^m$, which includes reproducible-goods settings, but it does not structurally enforce $\sum_i x_{ij} \leq 1$. For unit-supply auctions we learn this additional constraint using positive-excess penalties and report its failures. We do not apply clipping or rescaling as a post-processing feasibility repair.

Contributions. We give a sign-aware bounded-gradient construction and a constructive approximation theorem that survives its normalization, including revenue convergence and vanishing expected excess for represented feasible targets. We also identify a globally justified allocation-floor removal that leaves payments unchanged. Our empirical study then separates three questions: *representation*, using binary single-bidder types and finite-menu controls; *optimization*, using a fixed cost continuation in TEDI's reproducible-goods benchmark; and *compatibility*, using held-out unit-supply evaluation. The results include both successful regimes and failures: near-separable behavior at ten binary items, an approximate grand bundle at twenty, and low-average-excess models with non-negligible profile-level supply tails. Comparisons with published TEDI and GemNet values are outcome references, not matched-budget performance claims.

Full proofs, additional diagnostics, and reproduction instructions accompany the paper in the supplementary material.



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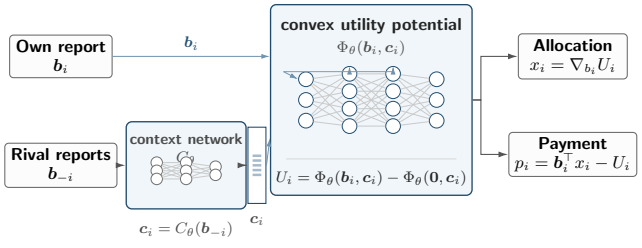


Figure 1: DolmaNet per-bidder architecture. Rival reports form context c_i; the convex utility potential yields allocation by own-report differentiation and payment by the envelope identity.

2 Setting and the Supply Relaxation

There are n bidders and m goods. Bidder i has a nonnegative additive type $t_i \in T_i \subseteq [0, \bar{t}]^m$. Each T_i is compact, convex, contains 0, and has nonempty interior. Write $T = \prod_i T_i$, and let $\mathcal{F}$ be a distribution on T from which training samples can be drawn. The incentive results are distribution-free. Absolute continuity of $\mathcal{F}$ is imposed only for the allocation and revenue approximation theorem.

An arbitrary report is $b_i \in T_i$. An allocation vector $x_i \in [0,1]^m$ has value $\langle t_i, x_i \rangle$, and utility after payment p_i is $\langle t_i, x_i \rangle - p_i$. Allocations may be divisible quantities or marginal probabilities for a risk-neutral bidder. In the latter interpretation, the incentive and participation statements refer to expected utility over allocation randomness, conditional on the type profile.

Incentives. A direct rule (x, p) is dominant-strategy incentive compatible (DSIC) if, for every i, t_i, b_i, b_{-i},

$$\langle t_i, x_i(t_i, b_{-i}) \rangle - p_i(t_i, b_{-i}) \geq \langle t_i, x_i(b_i, b_{-i}) \rangle - p_i(b_i, b_{-i}).$$

Ex-post individual rationality (IR) requires nonnegative truthful utility at every profile. We use a positive payment to mean a transfer to the seller.

Two distinct outcome constraints. The product space $\mathcal{X}_{\text{prod}} = \prod_i [0,1]^m$ enforces individual bounds. Unit supply additionally requires

$$\mathcal{X}_{\text{unit}} = \{x \in \mathcal{X}_{\text{prod}} : \sum_i x_{ij} \leq 1 \text{ for every } j\}.$$

DolmaNet maps reports into $\mathcal{X}_{\text{prod}}$ by construction. When it is used to learn a unit-supply auction, membership in $\mathcal{X}_{\text{unit}}$ is a separate constraint on every report profile. A finite training or test sample cannot establish that constraint everywhere.

For a candidate rule, define its revenue and overload statistics by

$$R_\theta = \mathbb{E}_{t \sim \mathcal{F}} \sum_i p_i^\theta(t), \tag{1}$$

$$e_j^\theta(b) = \left(\sum_i x_{ij}^\theta(b) - 1 \right)_+, \qquad V_\theta(b) = \sum_j e_j^\theta(b), \tag{2}$$

$$M_\theta(b) = \max_j e_j^\theta(b).$$

We distinguish expected excess, $\mathbb{E}V_\theta$, from a violation probability $\Pr(M_\theta > \epsilon)$ at an explicitly stated tolerance ϵ, and from worst-case excess. None is interchangeable with the others.

3 Rival-Conditioned Convex-Potential Auctions

3.1 One Utility, Derived Allocations and Payments

For each bidder, let $c_i = C_i(b_{-i})$ be an own-report-independent context. Define

$$U_i(b_i, b_{-i}) = \Phi_\theta(b_i, c_i) - \Phi_\theta(0, c_i), \tag{3}$$

$$x_i(b_i, b_{-i}) = \nabla_{b_i} \Phi_\theta(b_i, c_i), \tag{4}$$

$$p_i(b_i, b_{-i}) = \langle b_i, x_i(b_i, b_{-i}) \rangle - U_i(b_i, b_{-i}). \tag{5}$$

The context is fixed when taking the own-report derivative, but it remains trainable with respect to network parameters. Own-report independence must be a property of the forward computation, not an artificial stop-gradient applied to an own-dependent context.

Proposition 1 (Utility-potential implementation). *Fix the rival reports. Suppose U_i is convex, $U_i(0) = 0$, and a selected subgradient satisfies $x_i(b_i) \in \partial U_i(b_i) \cap [0,1]^m$ at every report. With payment $p_i(b_i) = \langle b_i, x_i(b_i) \rangle - U_i(b_i)$, truthful reporting is optimal, truthful utility is nonnegative, and payments are nonnegative. For a differentiable potential, use its gradient.*

Proof. For true type t_i and report b_i, the resulting utility is

$$U_i(b_i) + \langle t_i - b_i, x_i(b_i) \rangle \leq U_i(t_i)$$

by the subgradient inequality. Truthful reporting attains equality. Applying the same inequality at zero gives $U_i(t_i) \geq \langle t_i, x_i(0) \rangle \geq 0$. Applying it at t_i and evaluating at zero gives $0 \geq U_i(t_i) - \langle t_i, x_i(t_i) \rangle = -p_i(t_i)$. □

The proposition concerns incentive and participation constraints. Joint supply is not one of its conclusions. Under the product outcome constraint, the associated implicit menu has restricted conjugate price

$$P_i(q; b_{-i}) = \sup_{t_i \in T_i} \{\langle t_i, q \rangle - U_i(t_i, b_{-i})\}.$$

For $q = x_i(b_i, b_{-i})$, the subgradient inequality gives $P_i(q; b_{-i}) = p_i(b_i, b_{-i})$. Thus the gradient selects a supporting menu option without solving a separate optimization problem.

3.2 Contextual Input Convexity

For a fixed context c, use a partially input-convex network [1]:

$$\begin{aligned} z_1 &= \sigma(B_0(c) b_i + d_0(c)), \\ z_{\ell+1} &= \sigma(A_\ell(c) z_\ell + B_\ell(c) b_i + d_\ell(c)), \\ \Phi_\theta(b_i, c) &= \alpha(c)^\top z_L + \beta(c)^\top b_i + \delta(c). \end{aligned} \tag{6}$$

Here $A_\ell \geq 0$, $\alpha \geq 0$, and σ is convex and nondecreasing. Direct own-report matrices B_ℓ may be signed. These conditions imply own-type convexity by induction. The implementation uses $\sigma_\kappa(s) = \log(1 + e^{\kappa s})/\kappa$ with $\kappa > 0$, so $0 \leq \sigma_\kappa' \leq 1$. Context-dependent weights are generated by neural subnetworks; no convexity restriction is imposed on their dependence on rival reports.

A shared encoder can aggregate the masked rival set as

$$c_i = \rho_\theta \left(\left[\frac{\sum_{k \neq i} a_k \phi_\theta(b_k)}{\max(1, \sum_{k \neq i} a_k)}, \sum_{k \neq i} a_k \right] \right),$$

where $a_k \in \{0,1\}$ is a public active-bidder mask. Mean and count are both included. Shared parameters give bidder equivariance when no bidder-specific features are used. Public identities or bidder-specific networks can instead encode asymmetric roles. All expressiveness statements are relative to the chosen context representation and sharing constraints.

3.3 Sign-Aware Derivative Normalization

For coordinate j, propagate bounds on hidden derivatives, with elementwise extrema:

$$\begin{aligned} \underline{G}_{1,j} &= \min(B_{0:,j}, 0), \\ \overline{G}_{1,j} &= \max(B_{0:,j}, 0), \\ \underline{G}_{\ell+1,j} &= \min(A_\ell \underline{G}_{\ell,j} + B_{\ell:,j}, 0), \\ \overline{G}_{\ell+1,j} &= \max(A_\ell \overline{G}_{\ell,j} + B_{\ell:,j}, 0). \end{aligned} \tag{7}$$

Let $\underline{S}_j = \alpha^\top \underline{G}_{L,j}$ and $\overline{S}_j = \alpha^\top \overline{G}_{L,j}$ bound the nonlinear output derivative. Given $\eta_j(c) \geq 0$ and a numerical margin $0 \leq \mu < 1/2$, set

$$\begin{aligned} \gamma_j &= \frac{\max\{\eta_j + \overline{S}_j - \underline{S}_j, 1\}}{1 - 2\mu}, \\ \widetilde{B}_{\ell:,j} &= B_{\ell:,j}/\gamma_j, \\ \widetilde{\beta}_j &= \frac{\eta_j - \underline{S}_j}{\gamma_j} + \mu. \end{aligned} \tag{8}$$

All other weights remain unchanged. The normalization is inside the potential, before differentiation.

Lemma 1 (Global coordinate bounds). *The normalized potential is convex in its own input and, at every report,*

$$a_j(c) := \mu + \frac{\eta_j(c)}{\gamma_j(c)} \leq \partial_{b_{ij}} \Phi_\theta \leq \mu + \frac{\eta_j + \overline{S}_j - \underline{S}_j}{\gamma_j} \leq 1 - \mu.$$

Proof. Since $0 \leq \sigma' \leq 1$ and $A_\ell \geq 0$, induction gives the intervals in (7). Their endpoints are positively homogeneous in the direct paths of a fixed input coordinate. Dividing those paths by γ_j divides the endpoints by γ_j; adding $\widetilde{\beta}_j$ gives the stated bounds. Convexity is preserved because recurrent and output hidden weights remain nonnegative. This argument rescales interval bounds, not the old network's actual gradient: the new preactivations generally differ. □

Proposition 1 now applies for every parameter setting. Smoothness is needed only in the own input; the normalizer need not be smooth in rival reports or parameters. A positive margin restricts the represented class. The full approximation theorem uses $\mu = 0$, or a sequence of margins tending to zero, not a fixed positive margin.

Signed input paths permit substitution between items: increasing one item's value can reduce another item's allocation. Requiring every direct path to be nonnegative would rule out such mixed derivatives for our smooth activation. The supplement gives an example and interprets incentives using tangent planes.

3.4 Removing Free Allocation Floors

The affine part of a normalized utility contributes allocation but cancels from its envelope payment. A globally valid lower slope can therefore be removed without revenue loss.

Proposition 2 (Revenue-invariant floor removal). *Suppose $a_i(c_i) \geq 0$ is independent of the own report and $x_i(b_i, c_i) \geq a_i(c_i)$ everywhere. Define*

$$U_i^\flat(b_i, c_i) = U_i(b_i, c_i) - \langle a_i(c_i), b_i \rangle, \qquad x_i^\flat = x_i - a_i(c_i).$$

Then $U_i^\flat$ satisfies Proposition 1, its payment is exactly $p_i^\flat = p_i$ at every profile, and every item load weakly decreases.

Proof. Subtracting an affine function preserves convexity and zero normalization. The assumed lower bound gives $0 \leq x_i^\flat \leq x_i \leq 1$. The payment identity follows by cancellation: $\langle b_i, x_i - a_i \rangle - (U_i - \langle a_i, b_i \rangle) = p_i$. The load claim follows by summing nonnegative reductions across bidders. □

For Lemma 1, use $a_{ij} = \mu + \eta_j/\gamma_j$. Implementation requires retaining the *same* normalized nonlinear weights and γ_j, and replacing only the output coefficient by $-\underline{S}_j/\gamma_j$. Setting $\eta_j = 0$ and recomputing γ_j is a different operation and need not preserve payments. A sampled minimum, or the allocation at the zero type, is not a substitute for a global coordinatewise lower bound. Floor removal is optional and does not guarantee joint feasibility or redistribute unused supply.

3.5 Evaluation and Training Derivatives

For every bidder, compute its utility and its own-input gradient with rival reports fixed; then use (5). Independent bidder evaluations admit vectorization. Differentiating the sum of all bidder utilities with respect to the entire report profile is incorrect, because it includes rival-context derivatives. Training requires mixed derivatives through the allocation gradient; avoiding regret optimization does not make training first-derivative-only. All timing comparisons must measure the implemented full computation.

4 Learning Supply Feasibility

4.1 A Constraint-Relaxed Objective

For a batch of B profiles, the normalized unit-supply training objective is

$$\mathcal{L}(\theta) = \frac{1}{Bm} \sum_{k=1}^{B} \left[-\sum_i p_i^\theta(t^k) + \sum_j \{\lambda_1 e_j^\theta(t^k) + \lambda_2 (e_j^\theta(t^k))^2\} \right]. \tag{9}$$

The reported unit-supply runs use $(\lambda_1, \lambda_2) = (10, 50)$. Revenue and both positive-excess terms are averaged over items. Positive excess prevents under-allocation from cancelling over-allocation across profiles. The loss changes fitted parameters, not the structural incentive proof, and neither a small training loss nor a sampled risk threshold certifies joint feasibility.

The distinction between average excess and profile-level tails is quantitative. Let $e_j(t) = (\sum_i x_{ij}(t) - 1)_+$, $M(t) = \max_j e_j(t)$, and $\bar{e} = m^{-1} \sum_j \mathbb{E}[e_j]$. Then

$$\Pr(M > \tau) \leq \frac{\mathbb{E}[\sum_j e_j]}{\tau} = \frac{m\bar{e}}{\tau}. \tag{10}$$

Thus even an item-averaged mean excess of 10^{-4} gives only the loose bound $\Pr(M > 10^{-3}) \leq 0.5$ when $m = 5$. Controlling average violation is therefore not a substitute for measuring the tail event used at deployment.

4.2 Expressiveness and Feasible Targets

Fix continuous context maps C_i with compact images. A target is *represented* if its utility is jointly continuous in $(t_i, C_i(t_{-i}))$, convex in t_i, and admits allocation subgradients in $[0,1]^m$. Shared-network targets must admit the same shared factorization. The model class below is the union over finite widths and depths and universal context-parameter generators; it is not a claim about a fixed architecture or an information-losing fixed encoder.

Theorem 1 (Utility, allocation, and revenue approximation). *Assume $\mathcal{F}$ is absolutely continuous on T. For every represented, normalized DSIC and ex-post IR target (U^*, x^*, p^*), there is a sequence of zero-margin DolmaNet models such that, for every bidder,*

$$\|U_i^k - U_i^*\|_\infty \to 0,$$
$$\mathbb{E}_{\mathcal{F}}\|x_i^k - x_i^*\|_1 \to 0,$$
$$\mathbb{E}_{\mathcal{F}}|p_i^k - p_i^*| \to 0.$$

Consequently the increasing-capacity revenue supremum equals that of the represented continuous product-space DSIC/IR class. This is not a supremum restricted to exactly unit-feasible neural models.

Proof sketch. Use a finite net of *fixed* slopes in $[0,1]^m$, including zero. Their intercepts are negatives of the target's restricted convex conjugate and are continuous in context; thus no continuous selection of context-dependent subgradients is required. A nested soft maximum approximates their envelope with error at most $\tau \log Q$ and is realized by a signed-path PICNN. For this particular construction the propagated intervals recover the extrema of the affine slopes and the zero-margin normalization is the identity. Approximate the continuous intercepts with context generators and then normalize utility at zero. Uniform convergence of convex utilities gives gradient convergence at interior differentiability points; absolute continuity and bounded slopes give L^1 allocation convergence. The envelope identity yields payment convergence. The full construction, including strict-positivity closure, is in the supplementary material.

Corollary 1 (Approximation of feasible targets). *If the target in Theorem 1 also satisfies unit supply at every profile, the same sequence has $R_k \to R^*$ and*

$$\mathbb{E}V_k \le \sum_i \mathbb{E}\|x_i^k - x_i^*\|_1 \longrightarrow 0.$$

For every fixed $\epsilon > 0$, $\Pr(M_k > \epsilon) \to 0$.

Proof. Target feasibility implies

$$e_j^k = \left(\sum_i x_{ij}^k - 1\right)_+ \le \sum_i |x_{ij}^k - x_{ij}^*|.$$

Sum over items and take expectations. Since $M_k \le V_k$, Markov's inequality gives $\Pr(M_k > \epsilon) \le \mathbb{E}V_k/\epsilon$. Revenue convergence is part of Theorem 1. □

The corollary says that increasing capacity does not inherently force a nonzero expected-excess gap relative to a represented feasible target. It does *not* show that training finds the sequence, that a finite approximant is feasible everywhere, or that $\Pr(M_k > 0)$ tends to zero. The latter implication is false for general nonnegative sequences converging to zero in expectation. Nor does the theorem give a dimension-independent approximation rate.

The same convergence also controls the reproducible-goods objective used later. For fixed nonnegative costs c^p, c^d, define $D(x,p) = \sum_i p_i - c^p \sum_j \max_i x_{ij} - c^d \sum_{ij} x_{ij}$. Since $|\max_i x_{ij}^k - \max_i x_{ij}^*| \le \sum_i |x_{ij}^k - x_{ij}^*|$, Theorem 1 implies $\mathbb{E}|D(x^k,p^k) - D(x^*,p^*)| \to 0$. Thus the representation theorem applies to that product-space designer utility as well; it still says nothing about whether training finds the approximating sequence.

4.3 Interpreting Empirical Feasibility

Revenue of a rule with supply violations is not a revenue lower bound for the exactly feasible auction-design problem. The empirical gate is a validation selection rule, not a full-domain certificate. Mean positive excess, violation frequencies at stated tolerances, tail quantiles, and observed maxima describe different aspects of compatibility. Numerical regret tests can check an implementation but do not replace the convexity-based incentive proof.

5 Contextual Finite-Menu Controls

Our controls retain rival-only conditioning but replace the deep potential with a finite max-affine or log-sum-exp utility. These are established conditionally convex function classes [6]; their use here defines a comparison rather than an additional architectural novelty.

Hard menu. For context c_i, let $a_{iq}(c_i) \in [0,1]^m$ and $\pi_{iq}(c_i) \ge 0$ denote an allocation and a price, including a null option $(a_{i0}, \pi_{i0}) = (0,0)$. With scores

$$s_{iq}(b_i, c_i) = \langle a_{iq}(c_i), b_i\rangle - \pi_{iq}(c_i),$$

define $U_i^{\max}(b_i, c_i) = \max_q s_{iq}(b_i, c_i)$. At a tie, a fixed deterministic rule selects one maximizing option. Its slope is a valid subgradient; its price is exactly the envelope payment. The null option and nonnegative prices imply $U_i^{\max}(0, c_i) = 0$, so Proposition 1 applies. As with DolmaNet, independent own-optimal choices need not satisfy joint unit supply. This is the untransformed conditional-menu problem, not the compatibility guarantee of transformed GemNet [10].

Smooth menu. For a fixed temperature $\tau > 0$, define

$$G_{i,\tau}(b_i, c_i) = \tau \log \sum_q \exp\big(s_{iq}(b_i, c_i)/\tau\big),$$
$$U_{i,\tau}(b_i, c_i) = G_{i,\tau}(b_i, c_i) - G_{i,\tau}(0, c_i). \tag{11}$$

Let w_{iq} be the softmax of the scores divided by τ. The allocation is $x_i = \sum_q w_{iq} a_{iq}$, a convex combination of coordinate-bounded slopes, and payment is $p_i = \langle b_i, x_i\rangle - U_{i,\tau}$. Thus this is a truthful smooth utility rule in its own right, rather than merely a differentiable approximation to a hard deployed rule.

The distinction matters for payments. Define the entropy

$$H(w_i) = -\sum_q w_{iq} \log w_{iq}.$$

Then

$$p_i = \sum_q w_{iq}\pi_{iq} - \tau H(w_i) + G_{i,\tau}(0, c_i). \tag{12}$$

Consequently, the softmax-weighted original menu price is generally not the envelope payment at positive temperature. GemNet's soft training revenue and a normalized LSE utility therefore cannot be interchanged while claiming the same mechanism. Hard evaluation of a smoothly trained menu likewise changes the inference rule and must be identified separately.

Explicit-menu complexity on binary types. Ma et al. [7] use the binary-item example to illustrate the curse of discretization: all 2^m deterministic outcomes are needed by an exact explicit menu, although the optimal pricing rule has a compact description. For completeness, the corresponding statement in indirect-utility form is immediate. Fix $r \in (0, 1)$ and consider the separable posted-price mechanism that charges r for each purchased item.

Proposition 3 (Exponential exact menu size on the binary cube). *Let the type domain be $\{0, 1\}^m$. The posted-price mechanism*

$$x(v) = v, \qquad p(v) = r\|v\|_1$$

requires at least 2^m distinct allocation choices in any deterministic finite menu that implements the same allocation rule on every type. In contrast, the mechanism is induced by the separable convex indirect utility

$$U_r(v) = \sum_{j=1}^{m} (v_j - r)_+, \tag{13}$$

whose gradient equals v at every point of $\{0, 1\}^m$.

Proof. Every binary vector $v \in \{0, 1\}^m$ must select allocation $x(v) = v$. Distinct binary types therefore require distinct selected allocation vectors, so an exact explicit menu contains at least $|\{0, 1\}^m| = 2^m$ allocation choices, including the null allocation. Since $r \in (0, 1)$, each coordinate of $(v_j - r)_+$ has derivative 0 at $v_j = 0$ and derivative 1 at $v_j = 1$. Hence $\nabla U_r(v) = v$ on the binary domain, and the envelope payment is $v^\top v - U_r(v) = r\|v\|_1$. □

A softplus replacement of $(\cdot)_+$ gives the smooth approximation used by the constructive expressiveness argument. Thus explicit menu cardinality and potential-network capacity measure fundamentally different resources: the former enumerates choices, while the latter can compose exponentially many choices from a compact utility representation.

Scope of the controls. The controls reuse the rival-context building blocks, data distribution, and stated time budgets, but they do not isolate potential depth: optimizer, parameter count, and smooth-training/hard-evaluation differences remain. Here K denotes *learned non-null* options; the implementation appends a fixed null option, giving $K + 1$ choices. The binary counting argument concerns *exact allocation-rule reproduction*, not a lower bound on approximate expected revenue. Grand bundling illustrates this distinction in the supplement.

6 Empirical Results

DolmaNet headline tables use validation-only checkpoint selection followed by a held-out test set per setting, generated with seed 2026092601, $N = 131{,}072$, and batch size 256. Unless stated otherwise, we report the mean and sample standard deviation across training seeds 11, 23, 37. Published baseline numbers are labeled by source. The local GemNet reproduction is separate because it does not recover the published training frontier. Unless marked analytic, reference values are published estimates, not evaluations on our test set. Training trajectories and the checkpoint frontier instead use validation outputs.

	$B_{1,10}$	$B_{1,20}$
Itemwise pricing (analytic)	5.0000	10.0000
Grand bundle (analytic)	3.3125	6.9473
TEDI (published) [7]	4.9128	9.7927
DolmaNet	4.8755 ± 0.0202	6.8558 ± 0.0058
Contextual finite menu, $K = 32$	3.3431 ± 0.0288	6.8961 ± 0.0055
Contextual finite menu, $K = 300$	3.8161	7.0265
Contextual finite menu, $K = 1024$	3.9734	–

Table 1: Held-out test results on iid Bernoulli$(1/2)$ single-bidder settings. DolmaNet uses three training seeds; $K = 32$ uses two; $K = 300$ and $K = 1024$ use one. Multi-seed entries show mean ± sample standard deviation. Analytic rows are population expectations; purchase is selected at price indifference. K excludes the fixed null option.

6.1 Implicit menus and combinatorial type spaces

A single bidder removes joint supply from the problem and isolates the utility representation. Bernoulli values provide a particularly sharp test. Because this law is atomic, it lies outside the absolute-continuity assumption used for the allocation-convergence statement in Theorem 1; this experiment probes finite-model representation and optimization rather than that theorem: with m binary-valued items there are 2^m type vectors, while a finite-menu control contains K learned non-null options plus a fixed null option.

At $m = 10$, a post-freeze replay of the three selected checkpoints on the same held-out profile confirms nearly itemwise allocation. Averaged over items, $E[x_j \mid v_j = 1] = 0.99713 \pm 0.00030$ and $E[x_j \mid v_j = 0] = 0.000812 \pm 0.000114$ across training seeds. Payments are approximately linear in the number of active-valued items. This is consistent with separable item pricing. The exact allocation rule requires $2^{10} = 1024$ distinct choices including null; configured $K = 1024$ supplies 1025 choices and therefore meets that necessary counting condition, yet its held-out revenue is only 3.9734. Cardinality alone therefore does not explain the finite-budget gap, and the controls do not isolate optimization from representation or smooth-to-hard deployment.

At $m = 20$, the same post-freeze replay reveals a qualitatively different learned mechanism: on sampled binary types, all three seeds closely approximate a grand bundle with threshold 8 active-valued items. Mean per-item allocation is below 0.0012 at seven active items, jumps to 0.971–0.992 at eight, and is approximately one above the threshold; payments are near zero below threshold and about 7.9 above it. Among integer-priced bundles $k = 1, \ldots, 20$, price eight minimizes both allocation and payment mean absolute error for every seed. Relative to the exact price-8 bundle, mean allocation MAE is 0.00295 ± 0.00112, while the mean within-profile allocation standard deviation is only 6.1×10^{-4}. Figure 2 shows the contrast directly. The analytic price-8 bundle has population

revenue 6.9473, close to DolmaNet's 6.8558; itemwise pricing yields 10. With the reported margin, floor-removed allocations still permit the simple IR revenue ceiling 9.998, and a compact compatible soft-threshold potential attains analytic revenue approximately 9.7980 (supplement). Thus the observed bundle solution is a finite-budget training outcome, not a representational ceiling.

6.2 Reproducible goods: the TEDI benchmark

We also evaluate DolmaNet in the product-space environment introduced by Ma et al. [7], where U, B, and C denote uniform, Bernoulli, and correlated types, respectively, and the designer utility is

$$U_0 = \sum_i p_i - \sum_j \left(c^p \max_i x_{ij} + c^d \sum_i x_{ij} \right). \tag{14}$$

There is no hard unit-supply constraint here. In the direct full-cost diagnostic, optimization approaches a near-zero-allocation regime. We therefore use one fixed continuation rule across the benchmark: both cost coefficients in the *training loss* are ramped linearly from zero to their target values over the first 5,000 optimizer steps, while validation and checkpoint selection always use the full target costs.

Figure 3 illustrates the optimization behavior on $U^{1,0}_{3,2}$. The seed-11 direct full-cost diagnostic remains near zero over its recorded 30-second optimizer budget. The continuation diagnostic uses 90 seconds and first permits revenue to grow before trading it against the full target cost. These unequal-duration illustrative runs are not a matched causal ablation; the three-seed headline audit in Table 2 uses the separate 300-second protocol.

Across the final audit, DolmaNet's point estimate exceeds the published TEDI value on all four fixed-production-cost settings, while this remains a cross-paper outcome comparison rather than a compute-matched training comparison: $U^{1,0}_{3,2}$, $U^{1,0}_{3,10}$, $B^{1,0}_{3,10}$, and $C^{1,0}_{3,10}$. Both pure per-unit duplication-cost settings are below TEDI, substantially for $U^{0,.5}_{3,10}$ and slightly for $C^{0,.5}_{3,10}$. This is a cost-dependent pattern in the reported settings, not uniform dominance or an isolated architecture effect. In particular, the published separable benchmark remains higher on the binary fixed-production-cost case.

A contextual finite-menu control tests whether the same continuation suffices for the alternative parameterization. On $U^{1,0}_{3,10}$, applying the same 5,000-step cost continuation to hard deployed contextual menus gives held-out test designer utility 1.499 for $K = 32$ and 1.498 for $K = 300$ (one training seed each), versus 1.981 ± 0.005 for DolmaNet. Thus applying the same continuation does not close the observed gap. With one menu seed per size and different optimizers, this is a control diagnostic, not a causal estimate of the benefit of the utility representation.

6.3 Unit-supply auctions

For unit-supply experiments we select checkpoints by a fixed empirical validation gate. The "broad" gate requires mean positive item excess at most 10^{-4} and probability at most 1% that a profile has maximum item excess above 10^{-3}. This is an empirical gate, not an exact feasibility guarantee.

The two-bidder validation-gated selections retain useful revenue, and all six also satisfy the same empirical thresholds on the held-out test profile. The larger-bidder story is sharper: for 3×3 and 3×5, every seed has a gate-passing selection, but it is the untrained, step-zero initialization in every case. These near-null rules have mean revenues 2.9×10^{-4} and 1.5×10^{-4}. The corresponding raw revenue peaks reach 1.629 and 2.695, but fail the held-out 10^{-3} tail threshold on roughly 23–25% and 9.6–29% of profiles, respectively. Thus the obstruction is not truthfulness but the learned revenue–compatibility tradeoff. Figure 4 shows the same pattern over saved post-initialization validation checkpoints. The isolated high-revenue 4×3 gated star occurs in one seed; the other two selections again revert to initialization, so it is not a robust success. Full per-seed held-out tails, including p_{99} and observed maxima, are reported in the supplement.

GemNet reproduction audit. We use GemNet's published post-transformation values in Table 3. A separate implementation of the paper-described pipeline implemented the neural learner and Gurobi price transformation but did not recover the reported high-revenue/low-incompatibility training frontier. We therefore do not substitute our local numbers for the published baseline; the checkpoint and transformation audit is documented in the supplementary material.

7 Related Work and Limitations

Utility and menu representations. Rochet's convex-utility characterization underlies gradient allocation and envelope payment [9]. RochetNet uses a finite max-affine indirect utility for a single bidder [4], while parameterized max-affine and log-sum-exp networks approximate conditionally convex functions using context-dependent coefficients [6]. Generalized-convex representations likewise study function and gradient approximation for utility-based mechanism design [8]. DolmaNet's narrower contribution is a deep rival-conditioned potential with signed global derivative normalization, together with an approximation construction that survives that normalization. The binary menu example in Proposition 3 recasts TEDI's exact-compression observation in indirect-utility form; it is an illustration, not an independent novelty or approximate-revenue lower bound.

Multi-bidder neural mechanisms. RegretNet and attention-based extensions parameterize feasible allocations and optimize estimated regret [4, 5]; finite misreport search is an empirical lower bound on worst-case regret, not a global certificate. Lottery AMAs and AMenuNet instead learn within the affine-maximizer family [2, 3]. GemNet learns rival-conditioned finite menus using an incompatibility penalty and then applies offline price transformations with Lipschitz reasoning to obtain full-domain menu compatibility [10]. TEDI learns convex prices in allocation space and obtains choices by continuous utility maximization [7]. DolmaNet differs from both by learning indirect utility in type space and obtaining allocation directly by differentiation; neither product-space mechanism design nor supply penalties are themselves new.

Convex networks and scope. PICNNs provide the underlying convexity construction [1]; GroupMax and parameterized max/LSE networks give related convex approximation results [6, 11]. Our theorem is specific to represented continuous contexts, increasing capacity, sign-aware normalization, and an absolutely continuous

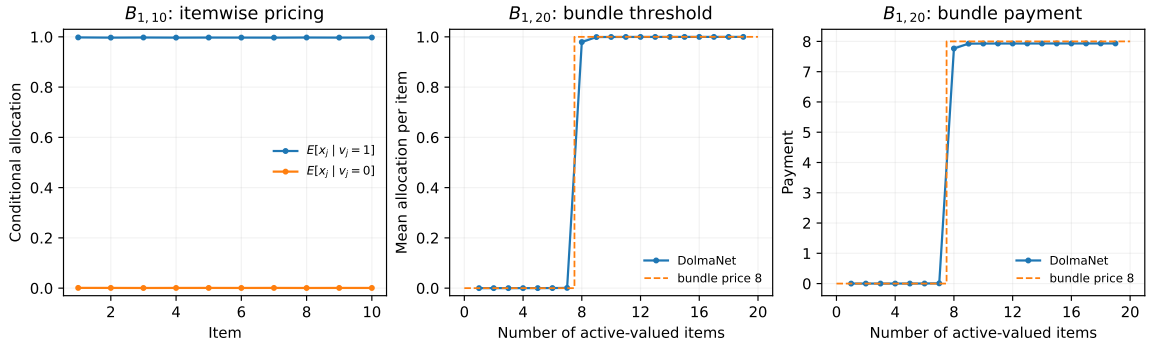


Figure 2: Frozen held-out mechanism structure. Left: the ten-item model allocates almost exactly itemwise. Middle/right: the twenty-item models instead learn an approximate grand bundle, with allocation and payment jumping at eight active-valued items. Solid DolmaNet curves are means over three training seeds; bands span those seeds. Dashed curves are the analytic price-8 bundle. These are post-selection diagnostics on the predeclared held-out profile, not additional model-selection criteria.

	$U^{1,0}_{3,2}$	$U^{1,0}_{3,10}$	$B^{1,0}_{3,10}$	$C^{1,0}_{3,10}$	$U^{0,.5}_{3,10}$	$C^{0,.5}_{3,10}$
TEDI [7]	0.2678	1.7285	5.8090	2.2400	2.1989	1.4101
MD-GemNet in TEDI [7]	0.2748	1.4814	1.3309†	1.7893	1.2848†	0.6350†
Separable-OPT [7]	0.2601	1.3019	6.2507	1.4981	1.8752	1.1894
DolmaNet	0.2830 ± .0010	1.9805 ± .0052	5.9246 ± .0432	2.6829 ± .0549	2.0186 ± .0575	1.3872 ± .0680

Table 2: Held-out test results on the six multi-player reproducible-goods settings reported by TEDI. Published TEDI, MD-GemNet, and Separable-OPT values are copied from Table 1 of its extended version and are not compute-matched to our runs. † preserves the source's indication of setting-specific hyperparameter tuning. DolmaNet entries are mean ± sample standard deviation over three training seeds, after validation-only checkpoint selection.

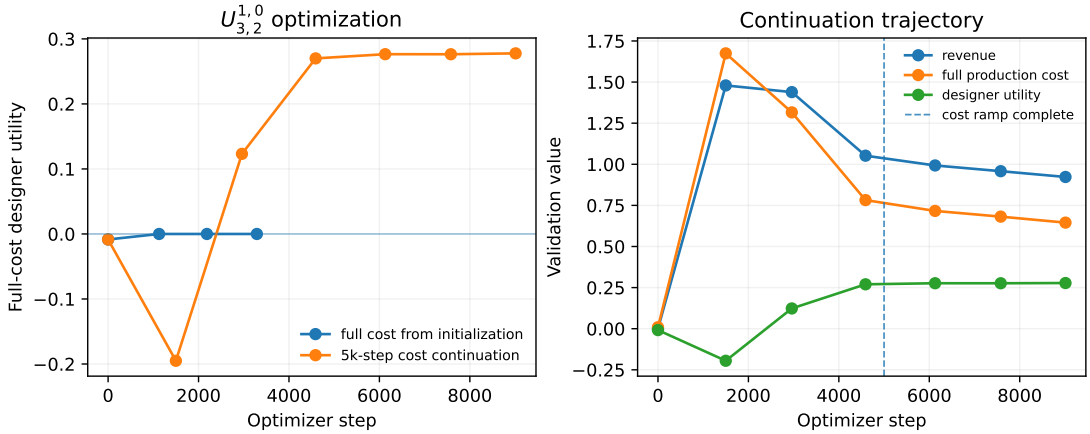


Figure 3: Seed-11 validation diagnostics on $U^{1,0}_{3,2}$: direct full-cost training (30 timed optimizer seconds) and cost continuation (90 seconds). The different horizons are shown explicitly; this is not a matched-duration ablation. Designer utility always uses the full target cost.

type law for allocation convergence. It does not imply a dimension-independent rate or reliable optimization.

The analysis is for nonnegative additive types. Shared unit supply is learned, not architectural, and no post-training feasibility repair is claimed. Randomized allocations satisfy the incentive statement in expected utility, not universal truthfulness for every realization. A fixed shared anonymous encoder need not represent asymmetric targets, and finite-precision implementations need not satisfy exact-arithmetic identities bit for bit. The empirical results therefore report both objective value and the constraint that remains learned, rather than treating raw revenue alone as a valid comparison.

	2 × 2 Add.	2 × 5 Add.	3 × 3 Add.	3 × 5 Add.
GemNet (published) [10]	0.8780	2.3100	1.6748	3.1237
Itemwise reserve 1/2 (analytic)	0.8333	2.0833	1.5938	2.6563
DolmaNet (validation-gated)	0.8664 ± 0.0008	2.1515 ± 0.0103	0.00029 ± 0.00005	0.00015 ± 0.00002
DolmaNet raw peak (diagnostic)	0.8714 ± 0.0037	2.2688 ± 0.0289	1.6288 ± 0.0070	2.6952 ± 0.0188

Table 3: Held-out test unit-supply audit on iid $U[0,1]$ additive settings. DolmaNet entries are mean ± sample standard deviation over three training seeds. "Raw peak" ignores the broad empirical feasibility gate and is shown only to expose the revenue–violation tradeoff. GemNet values are the published post-transformation results. The reserve-auction row is an analytic population reference, not the multi-item optimum. Gated 3×3 and 3×5 entries are untrained initialization checkpoints, not successful trained solutions.

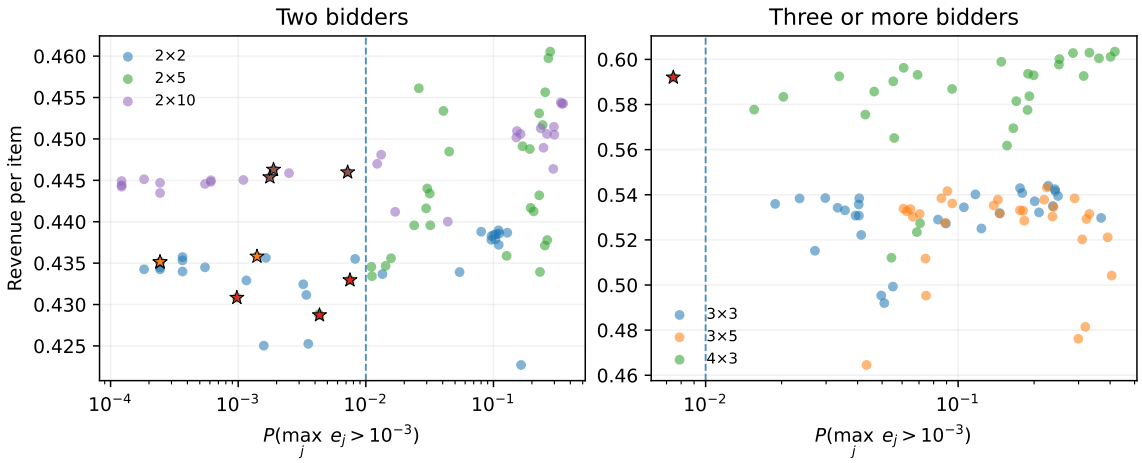


Figure 4: Revenue per item versus empirical profile-level supply violation over saved post-initialization validation checkpoints and three training seeds. Near-null initializations are omitted from this plot but included in Table 3. The dashed line is the 1% broad-gate threshold at excess tolerance 10^{-3}; stars mark validation-selected checkpoints that pass the broad gate.

8 Conclusion

DolmaNet learns rival-conditioned convex utilities with globally bounded gradients. Its signed-path normalization preserves structural bidderwise DSIC/IR and a constructive increasing-capacity approximation argument. Shared unit supply remains a separate learned constraint.

The experiments distinguish representation from optimization and compatibility. Ten-item binary models exhibit nearly itemwise pricing, while twenty-item models approximate a grand bundle despite a compact higher-revenue potential. Cost-continuation diagnostics illustrate optimization sensitivity, while comparisons with published TEDI values remain mixed and unmatched in compute. In unit-supply experiments, learned high-revenue rules can have substantial violation probabilities; gated three-bidder selections revert to initialization. Expressive truthful utilities therefore do not by themselves ensure successful optimization or shared-resource feasibility.

References

[1] Brandon Amos, Lei Xu, and J. Zico Kolter. 2017. Input Convex Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 70)*. 146–155.

[2] Michael J. Curry, Tuomas Sandholm, and John P. Dickerson. 2023. Differentiable Economics for Randomized Affine Maximizer Auctions. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*. 2633–2641. doi:10.24963/ijcai.2023/293

[3] Zhijian Duan, Haoran Sun, Yurong Chen, and Xiaotie Deng. 2023. A Scalable Neural Network for DSIC Affine Maximizer Auction Design. *Advances in Neural Information Processing Systems* 36 (2023), 56169–56185.

[4] Paul Dütting, Zhe Feng, Harikrishna Narasimhan, David C. Parkes, and Sai Srivatsa Ravindranath. 2024. Optimal Auctions through Deep Learning: Advances in Differentiable Economics. *J. ACM* 71, 1 (2024), 1–53. doi:10.1145/3630749

[5] Dmitry Ivanov, Iskander Safiulin, Igor Filippov, and Ksenia Balabaeva. 2022. Optimal-er Auctions through Attention. *Advances in Neural Information Processing Systems* 35 (2022), 34734–34747.

[6] Jinrae Kim and Youdan Kim. 2024. Parameterized Convex Universal Approximators for Decision-Making Problems. *IEEE Transactions on Neural Networks and Learning Systems* 35, 2 (2024), 2448–2459. doi:10.1109/TNNLS.2022.3190198

[7] Yunxuan Ma, Steven (Siqiang) Wang, Zhijian Duan, Yukun Cheng, and Xiaotie Deng. 2026. Learning Truthful Mechanisms without Discretization. In *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems*. 662–670. Extended version: arXiv:2506.22911v1. doi:10.65109/JTXL5273

[8] Moeen Nehzati. 2025. Universal Representation of Generalized Convex Functions and their Gradients. *arXiv preprint arXiv:2509.04477* (2025). doi:10.48550/arXiv.2509.04477

[9] Jean-Charles Rochet. 1987. A Necessary and Sufficient Condition for Rationalizability in a Quasi-Linear Context. *Journal of Mathematical Economics* 16, 2 (1987), 191–200.

[10] Tonghan Wang, Yanchen Jiang, and David C. Parkes. 2024. GemNet: Menu-Based, Strategy-Proof Multi-Bidder Auctions Through Deep Learning. In *Proceedings of the 25th ACM Conference on Economics and Computation*. Association for Computing Machinery, 1100. doi:10.1145/3670865.3673454

[11] Xavier Warin. 2024. The GroupMax Neural Network Approximation of Convex Functions. *IEEE Transactions on Neural Networks and Learning Systems* 35, 8 (2024), 11608–11612. doi:10.1109/TNNLS.2023.3240183